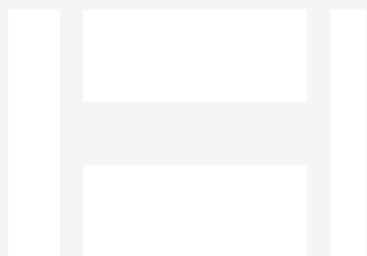


# Hwacheon Mega 95 for Industrial & General Manufacturing

Long-bed CNC lathe for extended shafts, rollers, rods, and structural round parts. This paper covers what the machine does, the industrial & general manufacturing parts we produce with it, the alloys we run, the tolerances and finishes industrial & general manufacturing buyers expect, and the documentation package.



# HWACHEON

**Rattle & Hum** · 16" OD × 120" L max turning · long-bed heavy-duty CNC lathe

## Executive summary

**Machine:** Hwacheon Mega 95 (shop nickname: *Rattle & Hum*). long-bed heavy-duty CNC lathe at B&R Productions, running from our New Waverly, Texas shop under an ISO 9001:2015 quality system.

**Application:** Long-bed CNC lathe for extended shafts, rollers, rods, and structural round parts for Industrial & General Manufacturing customers — Industrial equipment OEMs, custom machinery builders, pump and valve manufacturers outside oilfield, food and pharma equipment suppliers, mining and heavy-equipment operators, reverse-engineering customers, sustainment for out-of-production industrial equipment.

**Envelope:** 16" OD × 120" L max turning. **Tolerance:** ±0.0005" routine on critical features. **Traceability:** Material by heat and lot on every job.

---

## The machine, in context

Ten feet of turning length is a specialty capability. Long drive shafts, extended mandrels, rig-floor rollers, oil-and-gas structural rods — the parts that don't fit on a 4- or 6-foot lathe. Rattle & Hum lets us keep this class of work in-house instead of outsourcing the finish turning after roughing on a larger machine somewhere else.

### Full spec table

|                      |                                                                           |
|----------------------|---------------------------------------------------------------------------|
| Model                | Hwacheon Mega 95 ("Rattle & Hum")                                         |
| Type                 | Long-bed heavy-duty CNC turning center                                    |
| Max turning capacity | 16" outer diameter × 120" length                                          |
| Bed length           | 10-foot working length for extended parts                                 |
| Rigidity             | Cast-iron construction; steady-rest support for long slender work         |
| Tolerance capability | ±0.0005" routine on critical features; straightness held over full length |
| Materials            | Full oilfield alloy range — Inconel, Duplex, 17-4 PH, 4140/4340, Monel    |

---

## Who this serves in Industrial & General Manufacturing

**Typical buyer:** Industrial equipment OEMs, custom machinery builders, pump and valve manufacturers outside oilfield, food and pharma equipment suppliers, mining and heavy-equipment operators, reverse-engineering customers, sustainment for out-of-production industrial equipment.

**What's hard about industrial & general manufacturing work:** Legacy parts with no OEM support, one-off replacement components, small-lot production runs, materials that don't fit standard shop capabilities, and tight lead times on repair work where a down-machine is losing money by the day.

---

## Typical Industrial & General Manufacturing parts we produce on Rattle & Hum

- Long industrial drive shafts (up to 10 feet)
- Extended pump shafts and spindles
- Rollers and extended rollers
- Any industrial turned part that busts 4-foot class
- Long structural rods and threaded parts

Above list is representative; if your part isn't shown, call and ask — most oilfield / aerospace / defense / industrial turned or milled work fits somewhere in the shop.

---

## Alloys we run for Industrial & General Manufacturing

**Standard range:** Stainless (304, 316, 17-4 PH) · Tool steels · 4140 / 4340 · Brass · Bronze · Aluminum · Cast iron · Specialty alloys per customer requirement.

Industrial work covers the widest alloy range because the applications are the widest. Food and pharma often specify 316L or 17-4 PH; mining and heavy equipment specify tool steels and 4140/4340; pump manufacturers outside oilfield use everything from bronze bushings to stainless impellers. We work with the customer's spec sheet — if you tell us the alloy, we'll tell you honestly whether we can machine it and how quickly.

**What "we run these weekly" means:** we stock or reliably source the common alloys in the industry's typical spec range. Sharp tooling, proven feeds and speeds, and process control specific to each alloy family. Your part is not the one we're experimenting on.

---

## Tolerance and finish expectations in Industrial & General Manufacturing

±0.0005" routine on critical features. Industrial applications range from precision (pump seal areas, close-fit bushings) to structural (mining and heavy-equipment brackets); we adjust the inspection plan to what the part actually needs. Surface finish measured on parts where finish drives function.

---

## Documentation and quality workflow

Documentation calibrated to customer requirement. Material traceability standard for critical parts. Certificates of conformance issued with delivery. NDAs on request for proprietary designs or reverse-engineered legacy parts.

---

## Response time and emergency work

Down-machine and production-critical work prioritized. Lead times honestly quoted — no false promises. For legacy reverse-engineering, we typically confirm feasibility on the first phone call. Call (936) 291-7827.

---

## How we compare

Compared to sourcing a new OEM part: we're often faster and cheaper for one-offs and small runs, and for legacy parts where the OEM has moved on we're often the only realistic option. Compared to a cheapest-bidder job shop: we hold tolerance, document material, and don't ruin the part.

---

## Working with B&R Productions

New customer or existing, the process starts the same way: call (936) 291-7827 or send a print through the quote form. Prints are accepted as PDF, STEP, IGES, DWG, or DXF; if you only have a sample part, we reverse-engineer routinely. NDAs on request. ITAR-controlled prints handled under document control with prints staying in-shop.

Quotes typically come back within 24 hours for standard work, same-day for repeat customers on stocked materials, and near-immediately by phone for emergency and rig-down situations. Lead times are quoted honestly — we don't promise Wednesday and deliver next month.

Ready to talk about your Industrial & General Manufacturing work?

Send us a print or a sample photo and specs. Emergencies: call direct and say "rig-down."